

SL1620 DDR4 Core Module RDK

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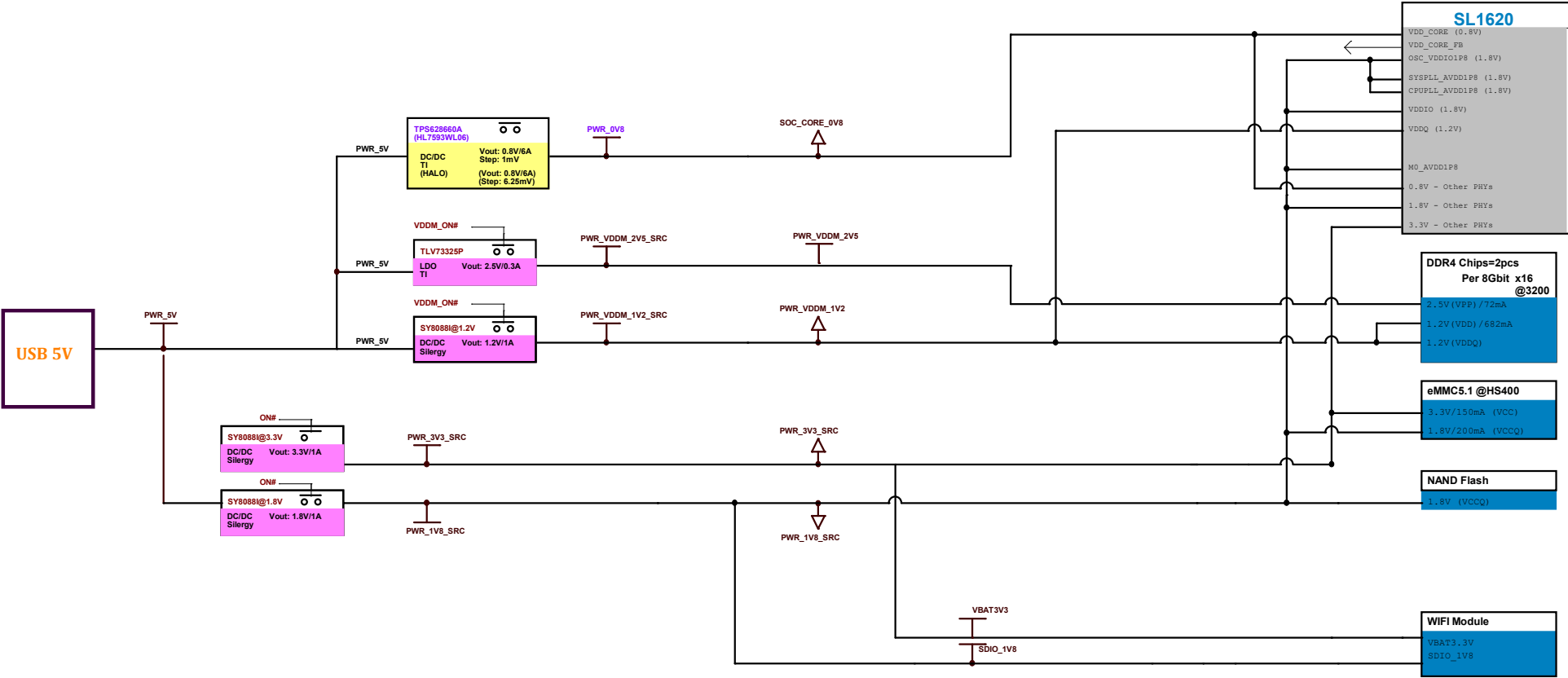
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PAGE#: DESCRIPTION

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Power Tree

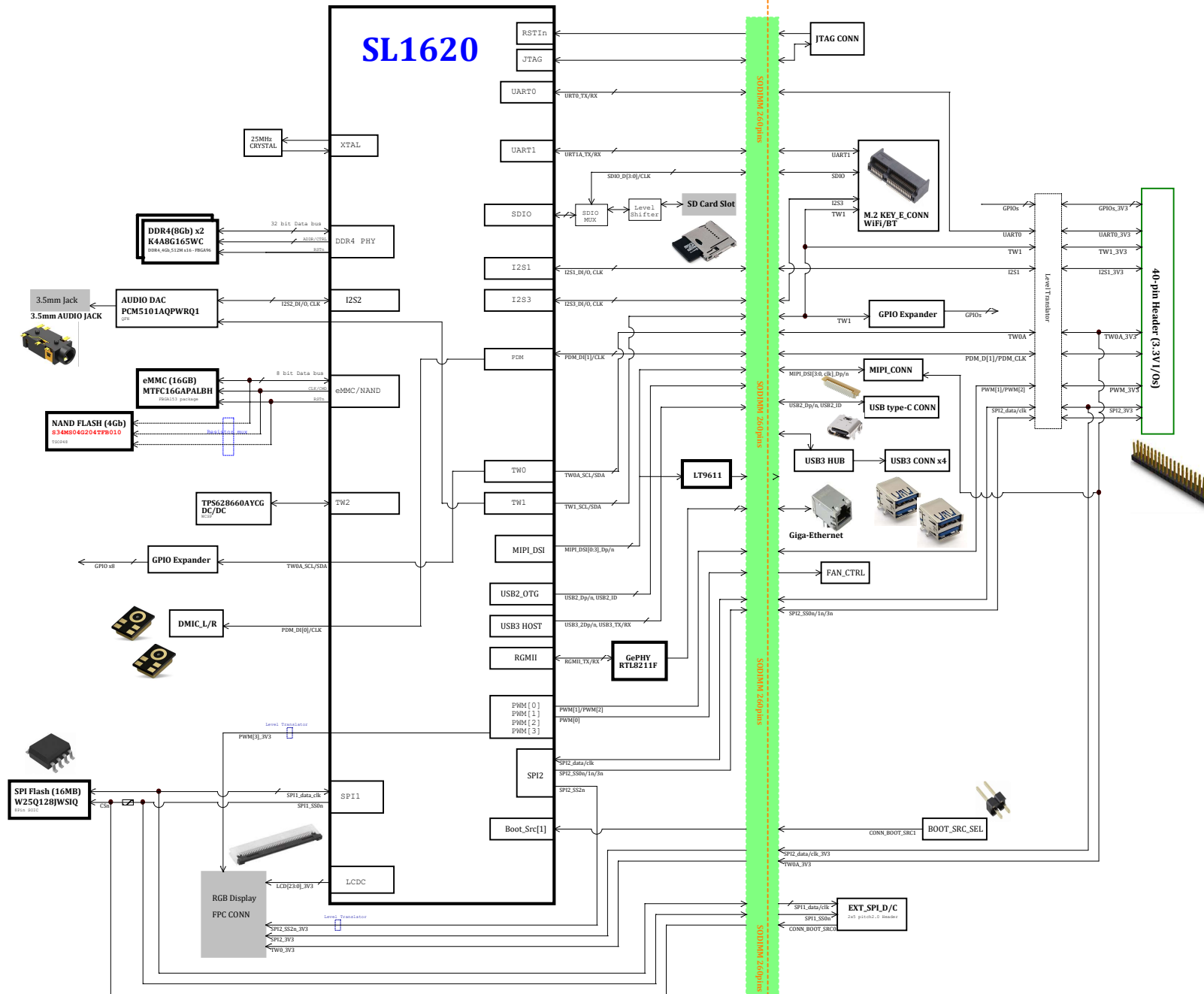


Revision History

REV	DATE	ORIGINATOR(S)	DESCRIPTION
A	01/13/2024	Iris Fan	Initial Release.
B	01/25/2024	Iris Fan	[Page.7] Adding PD resistors R1995~R2000 to unused I/Os of U83. [Page.16] Supply 1.8V power to U92 pin#18 (LDOO). [Page.16] Change R301 to Pull down U92 pin#12 (SCK) to enable internall PLL.
B	03/06/2024	Sheelan Zhu	[Page.7] 1. R41 changed to 33 ohm 2. add R55/R59 33 ohm, C35/C81 22pF for sdio sd/m2 clk [Page.8] 1. Del RGMII CLK OUT. NC 2. R65,R68, R74~R77 33 ohm on RGMII TX [Page.9] 1. R78/R79 33 ohm on I2S1 MCLK/BCLK 2. R84 33 ohm on LCD CLK [Page.13] R195 changed to 33 ohm [Page.14] add another I2C0 expander [Page.15] 1. R211,R212,R214,R231,R232,R234 changed to 33 ohm 2. 25MHz crystal for rtl8211 [Page.16] add DSI to HDMI converter [Page.19] R21 changed to 33 ohm
B	03/13/2024	Sheelan Zhu	[Page.7/8/9/19] 1. C128,C129,C132,C133 10pF DNS by default 2. C90 Stuffed by default [Page.22] add INA3221/INA220 to measure VCORE/VDDM_1V2/SOC_1V8/SOC_3V3 power
C	04/29/2024	Xiaomin Luo	[Page.18] 1. Replace U81 to directionless levelshifter TXB0104RGYR. 2. Change PWM[2] and GPIO[2] to 3.3V for 40 pin Header. 3. Remove USB-C_INTn signal.
D	06/21/2024	Xiaomin Luo	[Page 11] 1. Stuff R119. SW_STRP[3:1] = 3'b100 for Current sensor supporting. [Page 14] 1. Fix IOEXP ADDR/RST1 connection error. [Page.19] 1. Add 0.1uF cap between Q29 D/G. 2. Add 1Kohm. [Page.22] 1. Remove 1V8 and 3V3 here, they are in IO board. 2. Remove U4. 3. Measure VCORE, 1.2V, 2.5V with U9. [Page 21] 1. Add 4 standoffs for mounting heatsink

Core Module

I/O Board



MODE_0	MODE_1	MODE_2	MODE_3	MODE_4	STRAP	MISC
						CLK RCLKI RCLKO
						RST (PU) RSTIn
						(PU) POR_EN
						(PD) TEST_EN
						(PD) JTAG_SEL
						JTAG (PD) TRSTn
						(PD) TCK
TMS GPIO[0]	URT0B_TXD	KEY_COL2				TMS GPIO[1]
TDI GPIO[1]	URT0B_RXD	KEY_COL3	GPIO_TRIG[0]			TDI GPIO[2]
TDO GPIO[2]	PDMB_CLKIO	I2S2B_MCLK				TDO GPIO[3]
						UART
GPIO[40]	URT1A_TXD	KEY_ROW3				URT1_TXD
GPIO[39]	URT1A_RXD	KEY_ROW2	GPIO_TRIG[3]			URT1_RXD
						GPIO_A
GPIO[50]	RGMIIA_MDIO	PWM[1]				GPIO_A[0]
GPIO[49]	RGMIIA_MDC	PWM[2]				GPIO_A[1]
GPIO[48]		KEY_ROW5	URT1B_TXD			GPIO_A[2]
GPIO[47]		KEY_ROW4	URT1B_RXD			GPIO_A[3]

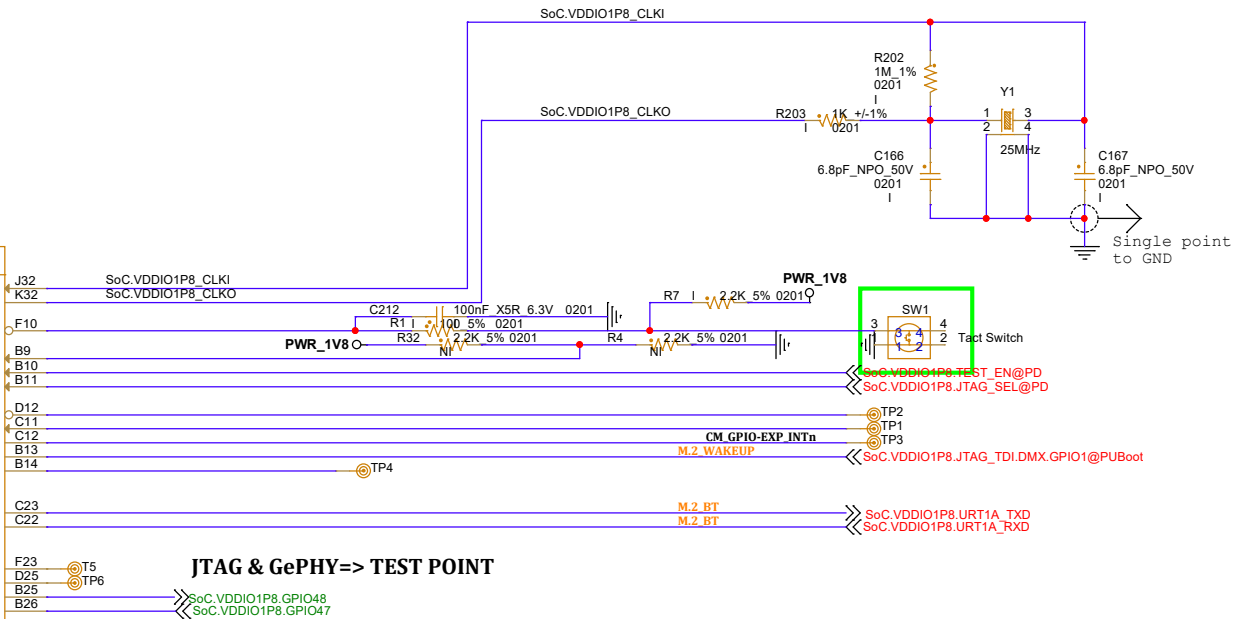
SL1620B0-BYOXSZZ-T000-T

MODE_0	MODE_1	MODE_2	MODE_3	MODE_4	STRAP	PWM
GPO[46]	PWM[0]	RGMIIA_PTP_PPS_O			cpuRstByps	PWM[0]
GPIO[45]	PWM[1]					PWM[1]
GPIO[44]	PWM[2]	KEY_COL1				PWM[2]
GPIO[43]	PWM[3]	IP_DCLS_SCLR				PWM[3]

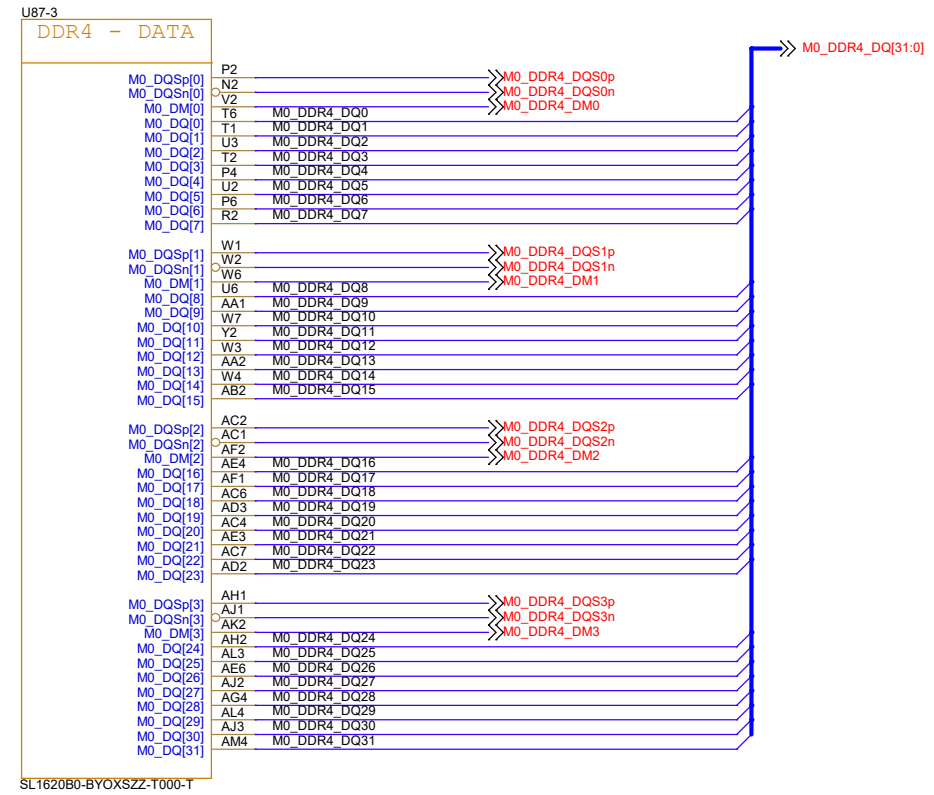
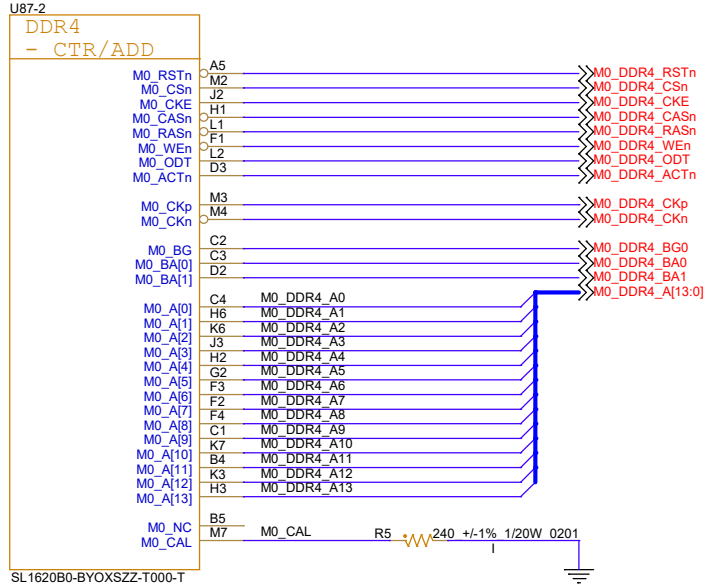
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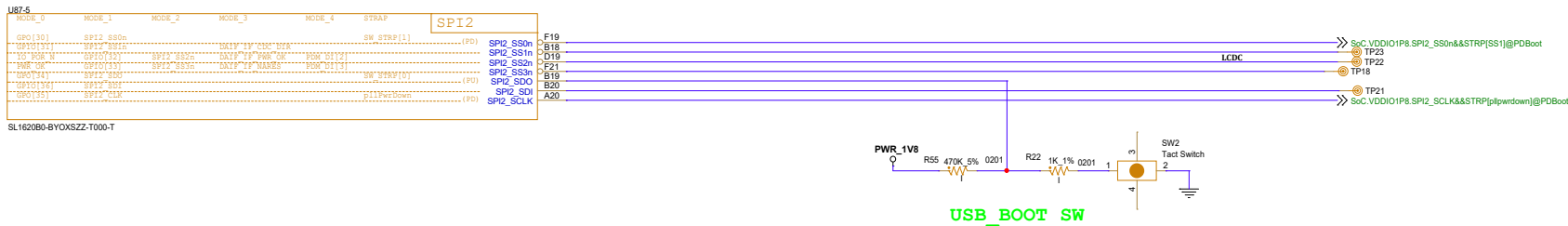
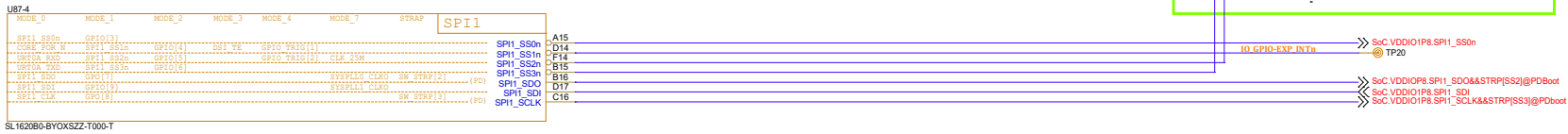
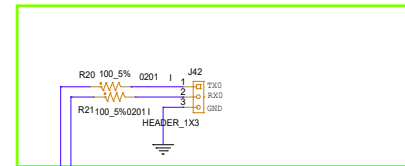
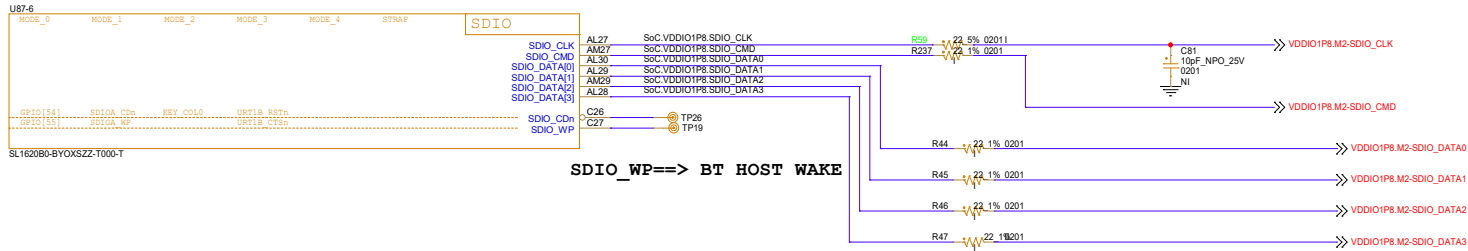
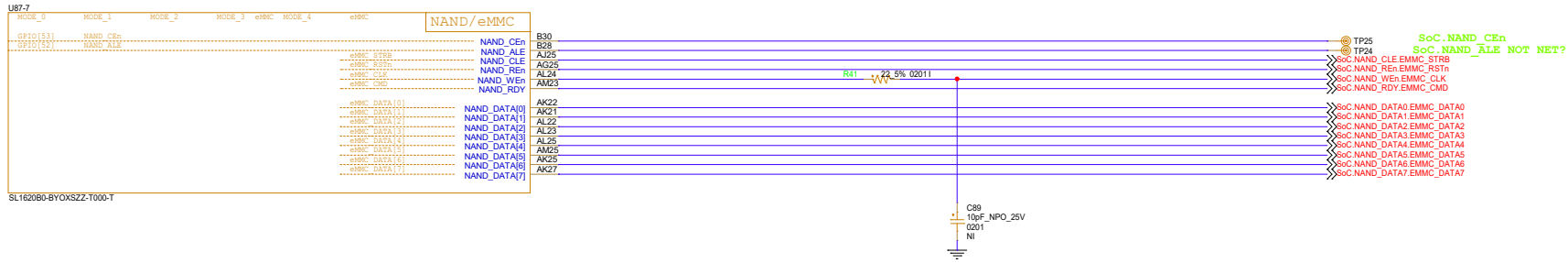
MODE_0	MODE_1	MODE_2	MODE_3	STRAP	TW
TW0A_SCL	GPIO[10]	URT0A_CTSn	I2S4_LRCK		TW0_SCL
TW0A_SDA	GPIO[11]	URT0A_RSTn	I2S4_BCLK		TW0_SDA
TW1_SCL	GPIO[12]		I2S4_DO		TW1_SCL
TW1_SDA	GPIO[13]	CLK_25M	I2S4_DI		TW1_SDA
GPIO[37]	TW2_SCL	KEY_ROW0	RGMITB_MDC		TW2_SCL
GPIO[38]	TW2_SDA	KEY_ROW1	RGMITB_MDIO		TW2_SDA
GPIO[41]	TW3A_SCL	URT1A_RSTn	IP_DCLS_N		TW3_SCL
GPIO[42]	TW3A_SDA	URT1A_CTSn			TW3_SDA

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FOXCONN	
Title	
05.SL1620 - MISC, PWM, TW	
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Rev	es





U87-10

MODE_0	MODE_1	MODE_2	MODE_3	MODE_4	STRAP	
						RGMII
						RGMII_CLK_OUT
GPIO_X[53]	RGMII_CLK_OUT					A30
						RGMII_TXC
GPIO_X[43]	RGMII_TXC					C30
GPIO_X[51]	RGMII_TXCTL					F30
GPIO_X[19]	RGMII_TXD[0]					D27
GPIO_X[20]	RGMII_TXD[1]					D31
GPIO_X[21]	RGMII_TXD[2]					D30
GPIO_X[22]	RGMII_TXD[3]					E31
						RGMII_RXC
GPIO_X[42]	RGMII_RXC					F28
GPIO_X[52]	RGMII_RXCTL					H30
GPIO_X[31]	RGMII_RXD[0]					F31
GPIO_X[32]	RGMII_RXD[1]					G31
GPIO_X[33]	RGMII_RXD[2]					G32
GPIO_X[41]	RGMII_RXD[3]					H31

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ETH RGMII DELETE

U87-15

MIPI_DSI

MIPI_DSI_CKp	AM9
MIPI_DSI_CKn	AL9
MIPI_DSI_D0p	AM6
MIPI_DSI_D0n	AL6
MIPI_DSI_D1p	AL8
MIPI_DSI_D1n	AM8
MIPI_DSI_D2p	AM11
MIPI_DSI_D2n	AM12
MIPI_DSI_D3p	AL12
MIPI_DSI_D3n	
MIPI_DSI_REXT	AM5

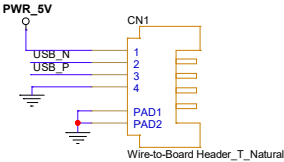
DVF120.MIPI_DSI_REXT

R73

200 1% 0201

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USB 5V Source



U87-13

MODE_0	MODE_1	MODE_2	MODE_3	STRAP	
					USB2
					USB2_Dp
					USB2_Dn
					USB2_ID
					USB2_VBUS
					USB2_DRV_VBUS
					USB2_REXT
					AL19
					AL18
					F25
					AL20

USB2_Dp

USB2_Dn

USB2_ID

USB2_VBUS

USB2_DRV_VBUS

USB2_REXT

AM19

USB_P

AM18

USB_N

AL19

AL18

F25

AL20

R94

20K 5% 0201

R90

20K 1% 0201

R96

200 1% 0201

R521

2.2K 5% 0201

NI

SoC.VDDIO1P8.GPO51&&STRAP[p1lByps]@PDboot

USB2_IDPIN_CTRL

L Host mode

H Device mode

PWR_5V

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U87-14

USB3

USB3_TXp	AL14
USB3_TXn	AM14
USB3_RXp	AL17
USB3_RXn	AM17
USB3_Dp	AL16
USB3_Dn	AL15
USB3_ID	AG16
USB3_VBUS	AJ16
USB3_REXT	AG12

R99

200 1% 0201

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U87-11						I2S/PDM	
MODE_0	MODE_1	MODE_2	MODE_3	MODE_4	STRAP		
GPIO[17]	I2S1_MCLK		DAIF IF CLK IF			I2S1_MCLK	M32
GPIO[15]	I2S1_BCLK		DAIF IF CLK DAT AD			I2S1_BCLK	M31
GPIO[14]	I2S1_LCKR		DAIF IF CLK DAT AD			I2S1_LCKR	L31
GPIO[14]	I2S1_DO		DAIF IF CLK DAT DA			I2S1_DO	M26
GPIO[18]	I2S1_DI		DAIF IF CLK DA	APLL1_CLK0	CODE SEC[0] (PD)	I2S1_DI	M27
GPIO[20]	I2S2_BCLK	URTOB RTSR	DAIF IF EN AD			I2S2_BCLK	P27
GPIO[19]	I2S2_LCKR	URTOB CTSR	DAIF IF DAT AD			I2S2_LCKR	N31
GPIO[22]	I2S2_DO		DAIF IF EN DA		CODE SEC[1] (PD)	I2S2_DO	P30
GPIO[22]	I2S2_DI		DAIF IF DAT DA	VREF1_CLK0	legacy boot (PD)	I2S2_DI	R30
GPIO[27]	I2S3_BCLK					I2S3_BCLK	T31
GPIO[26]	I2S3_LCKR					I2S3_LCKR	R31
GPIO[28]	I2S3_DO			APLL1_CLK0	legacy boot (PD)	I2S3_DO	R32
GPIO[29]	I2S3_DI			VREF1_CLK0		I2S3_DI	U32
GPIO[25]	PDM_CLKIO	I2S2A_MCLK				PDM_CLKIO	T26
GPIO[24]	PDM_DI[0]					PDM_DI[0]	T27
GPIO[23]	PDM_DI[1]					PDM_DI[1]	R29
					(PD) FORCE_MUTE		A10

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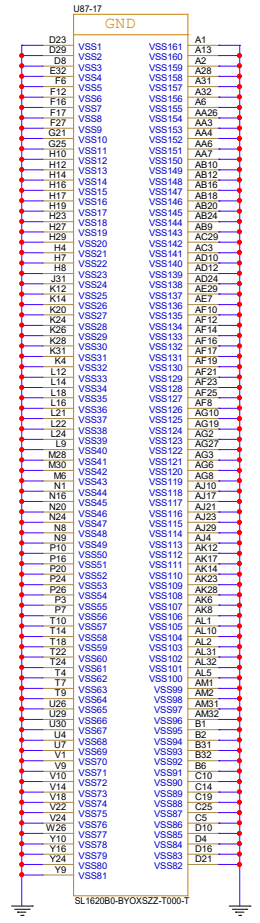
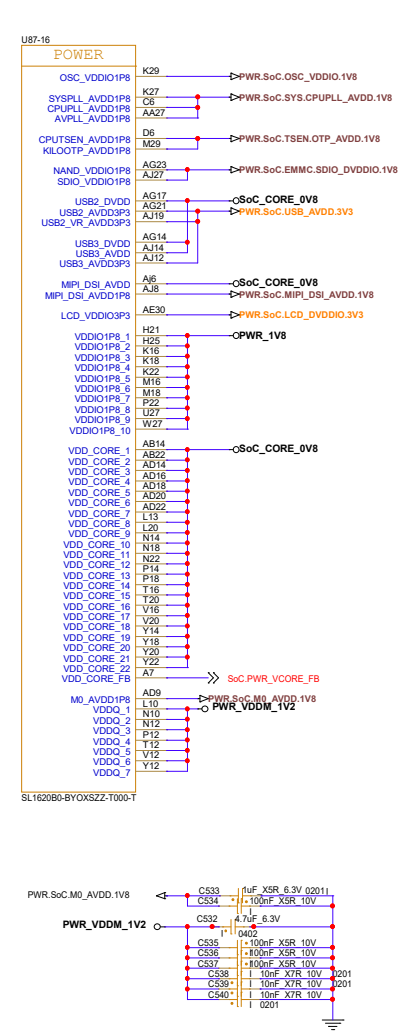
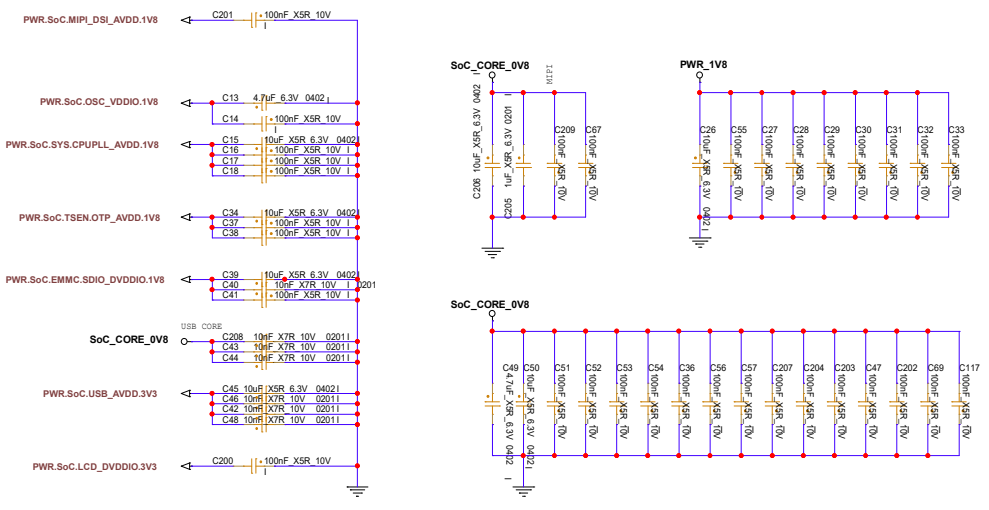
STRAP PIN

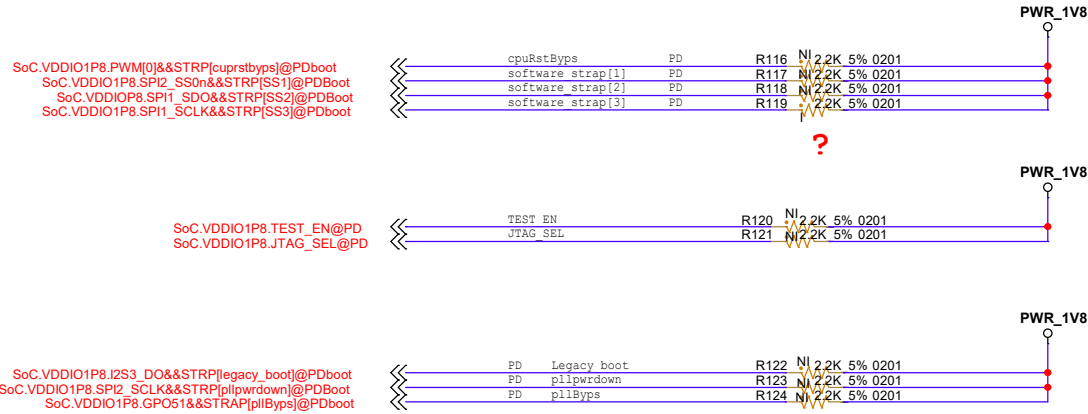
M2.BT PCM PIN

U87-12						LCD	
MODE_0	MODE_1	MODE_2	MODE_3	MODE_4	STRAP		
GPIO_X[12]	LCD D[0]	KEY_ROW6	TW3B_SCL			LCD D[0]	U31
GPIO_X[13]	LCD D[1]	REV_ROW7	TW3B_SDA			LCD D[1]	V31
GPIO[98]	LCD D[2]					LCD D[2]	W29
GPIO[97]	LCD D[3]					LCD D[3]	W30
GPIO[98]	LCD D[4]					LCD D[4]	Y32
GPIO[99]	LCD D[5]					LCD D[5]	Y31
GPIO[98]	LCD D[6]					LCD D[6]	AA31
GPIO_X[10]	LCD D[7]	REV_COL6	I2S3_DO			LCD D[7]	AA30
GPIO_X[11]	LCD D[8]	REV_COL7	I2S3_DI			LCD D[8]	AA29
GPIO[62]	LCD D[10]					LCD D[10]	AB32
GPIO[63]	LCD D[11]					LCD D[11]	AB31
GPIO[64]	LCD D[12]					LCD D[12]	AC30
GPIO[65]	LCD D[13]					LCD D[13]	AC27
GPIO[66]	LCD D[14]					LCD D[14]	AC26
GPIO[67]	LCD D[15]					LCD D[15]	AD30
GPIO_X[15]	LCD D[16]	REV_ROW8	I2S3_LCKR			LCD D[16]	AD31
GPIO_X[16]	LCD D[17]	REV_ROW9	I2S3_BCLK			LCD D[17]	AE32
GPIO[68]	LCD D[18]					LCD D[18]	AE31
GPIO[69]	LCD D[19]					LCD D[19]	AE27
GPIO[70]	LCD D[20]					LCD D[20]	AE26
GPIO[71]	LCD D[21]					LCD D[21]	AF31
GPIO_X[40]	LCD D[22]					LCD D[22]	AG31
GPIO_X[14]	LCD D[23]					LCD D[23]	AG29
GPIO_X[15]	LCD_CLK					LCD_CLK	AG32
GPIO_X[16]	LCD_GPIO[0]					LCD_GPIO[0]	AG30
GPIO_X[17]	LCD_GPIO[1]					LCD_GPIO[1]	AJ30
GPIO_X[18]	LCD_GPIO[2]					LCD_GPIO[2]	AJ31
GPIO_X[19]	LCD_GPIO[3]	REV_COL4	SDIOB_CSD			LCD_GPIO[3]	AK31
GPIO_X[11]	LCD_GPIO[4]	REV_COL5	SDIOB_WP			LCD_GPIO[4]	AK32

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LCD DELETE





TEST_EN 0: In function mode or JTAG mode, no SCAN mode. (Default)
1: In test mode, particular in SCAN mode.

JTAG_SEL 0: Select ICE debug mode (Default)
1: Select BSCAN TAP mode

POR_EN 0: External POR
1: Internal POR (Default)

legacy_boot 0: 2ms (Default)
1: 20ms

SW_STRAP[0] 0: Boot from USB
1: Boot from the device selected by Boot_SRC (Default)

software_strap[1]
Default:-0

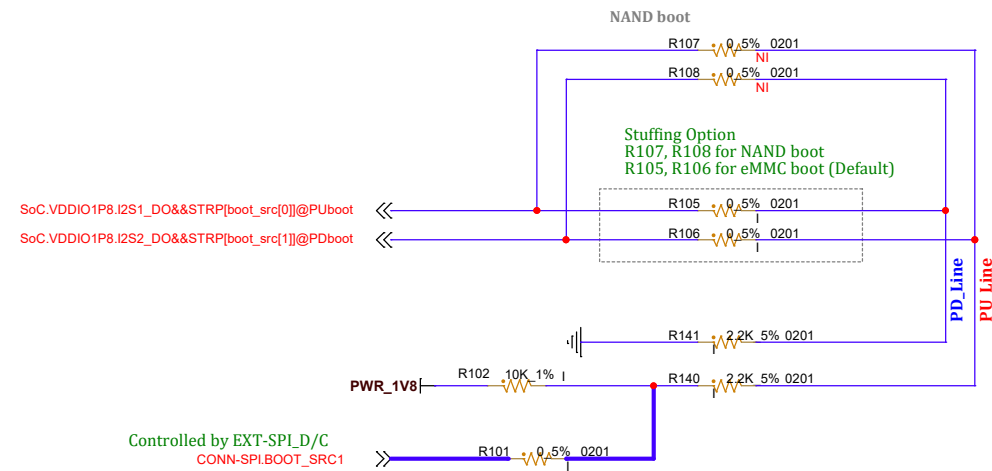
software_strap[2]
Default:-0

software_strap[3]
Default:-0

cpuRstByps 0: Enable reset logic inside cpu partition (Default)
1: Bypass reset logic inside cpu partition

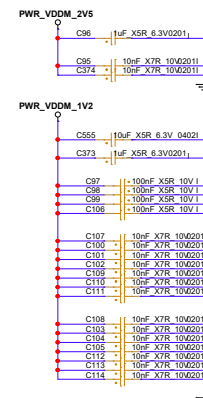
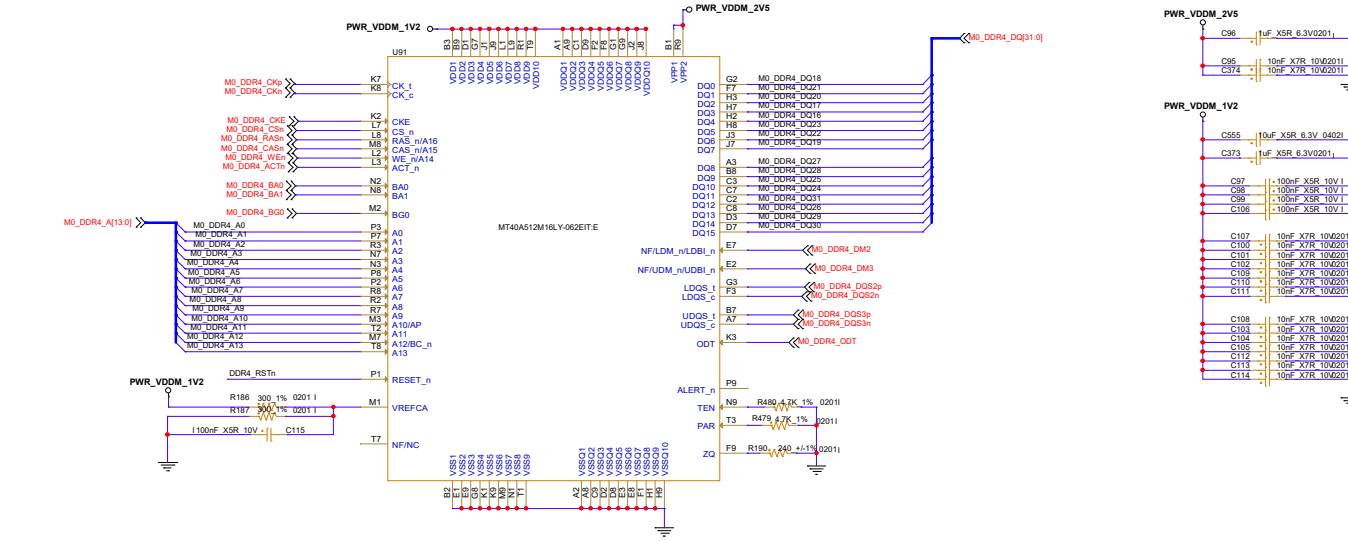
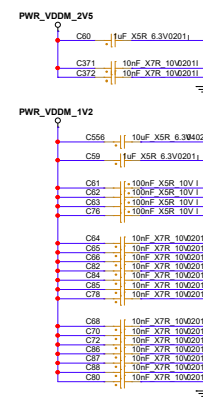
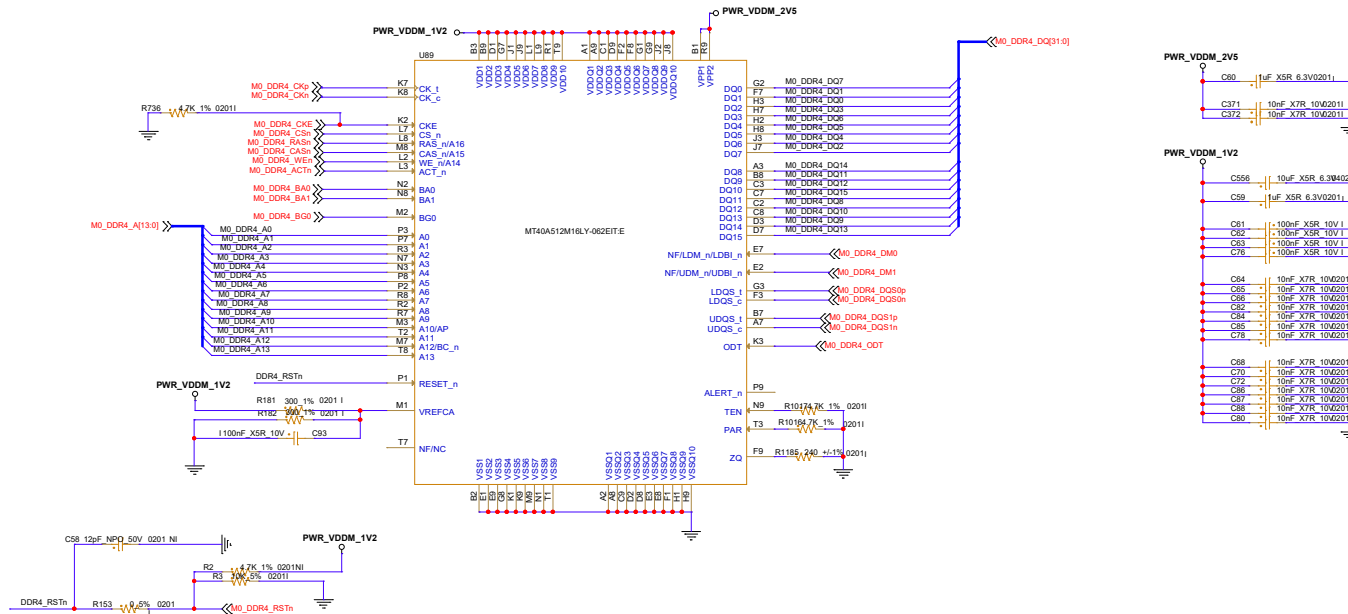
pllPwrDown 0: Power up SYS/MEM/CPU PLL (Default)
1: Power down SYS/MEM/CPU PLL

pllByps 0: No Bypass (Default)
1: All PLL bypassed

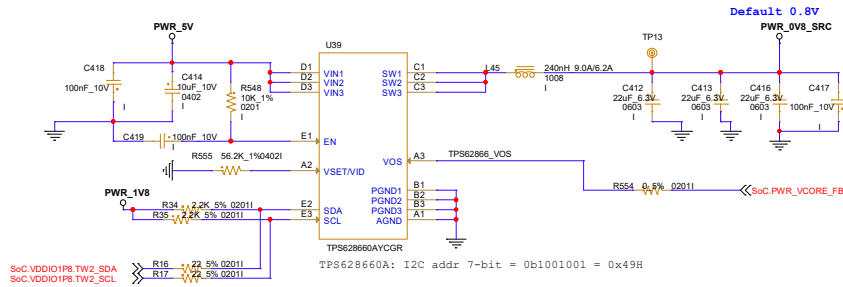


BOOT STRAP OPTION

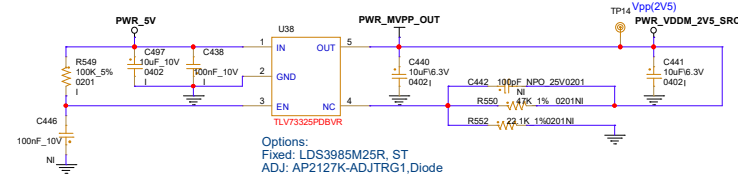
BootSrc[1:0]	TYPE
2'b00	SPI-Secure Boot
2'b01	ROM boot from NAND default
2'b10	ROM boot from EMMC
2'b11	SPI-Clear Boot
SPI2_SDO = 0 USB boot	



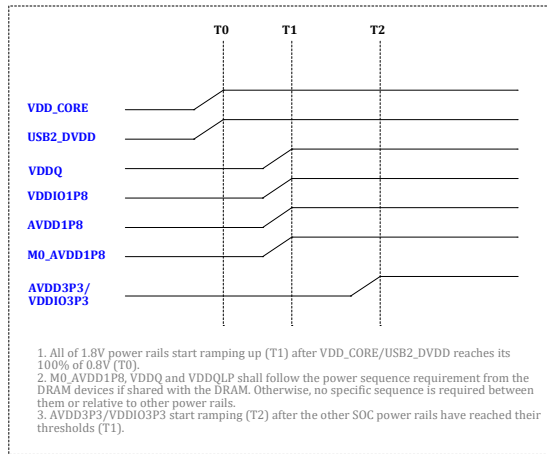
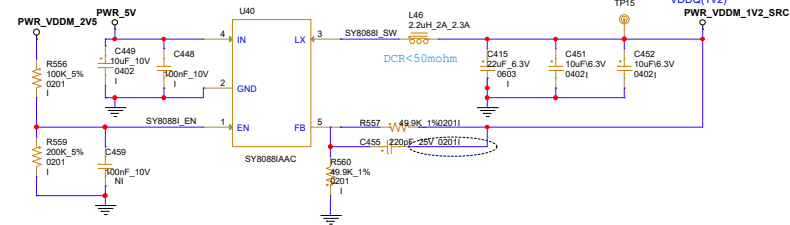
VDD_CORE POWER - 6A/5mV Step (Default)



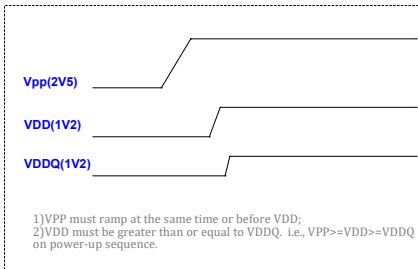
2.5V/0.3A for DDR-DEVICES;



1.2V/1A for DDR-SOC + DDR-DEVICES;

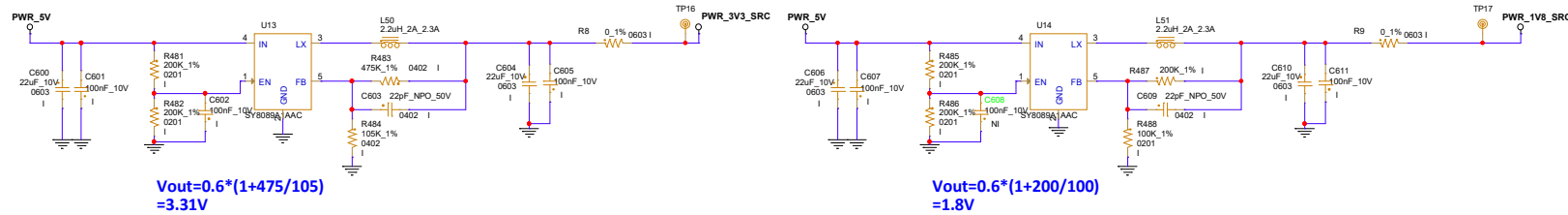


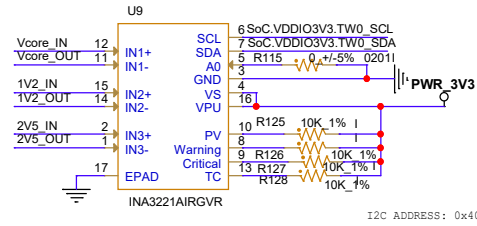
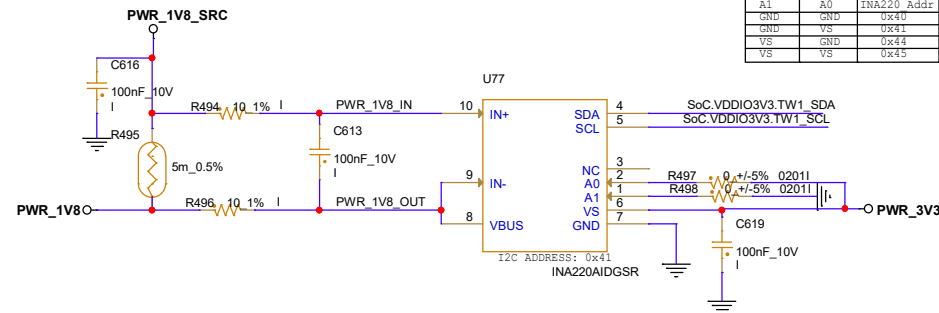
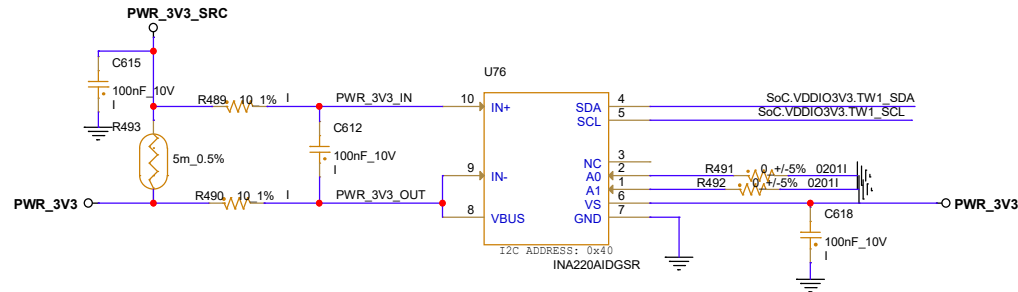
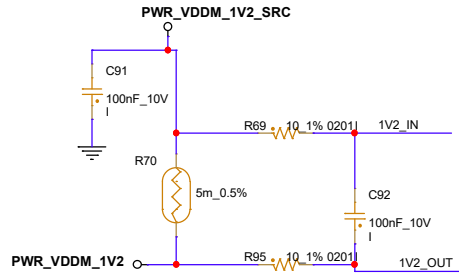
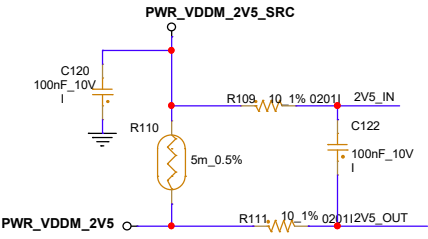
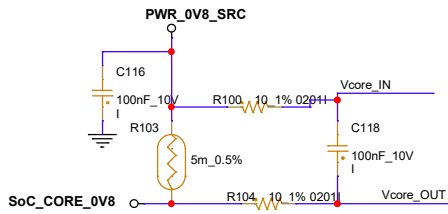
SL1620 Recommended Power Sequence



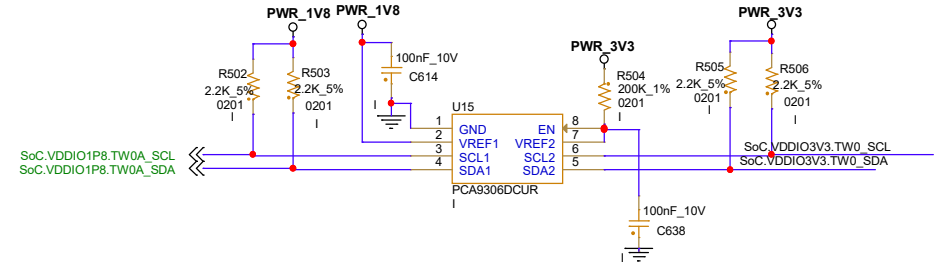
Power-Up Sequence For DDR4 DEVICE

ADD PWR_3V3&1V8

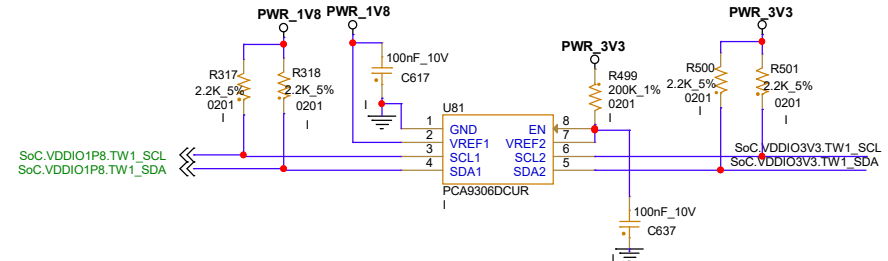




1V8_TW0 I2C => 3V3_TW0 I2C



1V8_TW1 I2C => 3V3_TW1 I2C



Title			
16 Current-Measurement			
Size	Document Number	Rev	
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Date:	Monday, January 20, 2025	Sheet	16 of 16